

SureClinical™
Collaborative Cloud Applications for Health Sciences

Evaluating SharePoint-based eTMF Solutions

Collaborative cloud computing enables organizations to create, digitally sign and complete documents in the cloud, eliminating paper at the point of origin. SharePoint-based solutions are costly, complex, difficult to use and are based on a 1990s-architected legacy workgroup platform that has fallen behind and has increasingly declining rates of user acceptance and adoption. Lacking support for basic features like mobile document viewing and E-signing, business process management and integrated FDA 21 CFR Part 11 and HIPAA compliance, Microsoft SharePoint is the dark cloud on the cloud computing horizon, threatening the success and ROI of BioPharma technology investments. [7]

© Copyright 2013, 2014 SureClinical Inc. http://www.SureClinical.com
SureClinical eTMF® is a registered trademark. SureClinical eTMF technologies are patent pending. Visit http://www.SureClinical.com for info.
SureClinical™, SureClinical, SurePortal, SureMailbox, SureDesigner, SureWorkflow, SureEsign, SureQuality are trademarks of SureClinical Inc,
No part of this document may be reproduced, stored in, or introduced into a retrieval system, or transmitted in any form or by any means including electronic, mechanical, photocopy, recording, or otherwise, or for any purpose, without the express written permission of SureClinical Inc.
This document is for discussion purposes only. SureClinical makes no warranties, express or implied, in this document. SureClinical cannot guarantee the accuracy of any information presented after the date of publication. This is not a commitment by SureClinical to deliver any product or service mentioned herein. All material subject to change. Implementing companies should consult with an independent advisor to ensure organization specific compliance requirements. The names of products mentioned herein may be the trademarks of their respective owners. SureClinical may have patents, patent applications, trademarks, copyrights, or other intellectual property rights covering subject matter in this document. Except as expressly provided in any written license agreement from SureClinical, the furnishing of this document does not give you any license to these patents, trademarks, copyrights, or other intellectual property.

Contents

Introduction	3
Collaborative Investigator Cloud Portals for Enhanced Efficiency	4
The Problems with Microsoft SharePoint	4
Requirements for Collaborative Cloud Portals for Regulatory Documents	4
The SharePoint Legacy	5
Evaluation of SharePoint as a Collaborative Cloud Platform for Reg Docs	5
SharePoint – Difficulties in Sharing Content Outside of the Organization	6
Evaluation Matrix: Cloud-based Collaborative eTMF Requirements vs. SharePoint	6
Why SharePoint Falls Short in BioPharma Extraprise Cloud Collaboration	10
SharePoint – Closed, Proprietary, Not Standards-based	11
SharePoint Applications: Patchwork of Custom Point Solutions	12
Sureclinical's Approach – What Sets Us Apart	12
Conclusion	14
References and Resources	15

About the Author

Zachariah Schmidt, BS Electronics/Computer Science, MBA
SureClinical

Headquartered in USA, SureClinical provides collaborative cloud application solutions that automate the regulatory document lifecycle for the Life Sciences and Healthcare industries. Prior to founding SureClinical, Mr. Schmidt led product development and marketing efforts at leading software firms Siebel, Oracle, and Adobe. Zack founded the OASIS eTMF Standards initiative, a global standards initiative to define machine-readable formats for clinical trial master file content interoperability and data exchange. He serves on the Board of Directors of CareLex, a not-for-profit enterprise involved in BioInformatics Research and Development.



Collaborative Cloud Portals vs. SharePoint-based eTMF Solutions

SharePoint - Proprietary Legacy Technologies That Limit Enterprise Flexibility

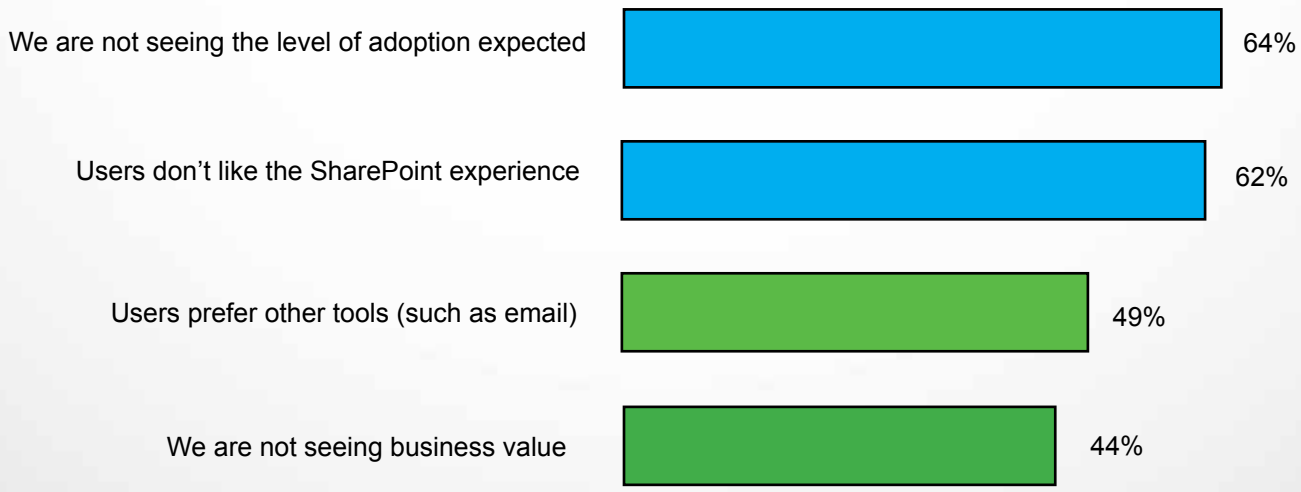
Collaborative cloud computing enables organizations to create, digitally sign and complete documents in the cloud, eliminating paper at the point of origin. Cloud applications are typically accessible across desktop, web and mobile devices. As a result of the significant cost and time savings and effective data management offered by collaborative cloud computing, user adoption is increasing rapidly in the enterprise.

On the other hand, SharePoint-based solutions are costly, complex, difficult to use and are based on a 1990s-architected legacy workgroup platform that has fallen behind. Instead of using open standards, the latest SharePoint 2013 platform is based on proprietary technologies that limit collaboration, integration and enterprise application flexibility, creating both technology investment and BioPharma compliance risks.

Introduction

In the fast-paced BioPharma business environment, collaborative cloud computing is gaining traction, providing enterprises with a means to improve operational efficiency and accelerate time to market. *Extraprise* collaboration – or the ability to collaborate with groups within the organization as well as with firms outside of the organization, including partners, customers and vendors - is now a fundamental business capability that should form the foundation of your clinical trial eTMF content management system.

Survey: In what ways is SharePoint unable to meet IT or business management expectations?



Source: Forrester Research Survey, October 2013

Collaborative processes play a role in almost every part of your business – from clinical trial research and development and clinical trial study startup to product marketing. The applications and technology infrastructure you use are essential to leverage and integrate the work of your entire internal and external clinical trial team, including sponsors, CROs, investigators, academic research organizations and other clinical trial partners. Collaborative cloud computing is at the center of a movement in the software industry to build functionally rich and vertically-focused solutions that can extend the power of enterprise computing beyond the firewall to include external business partners, in what is commonly referred to as **Extraprise** cloud computing.

Copyright 2013, 2014 SureClinical Inc.

Collaborative Investigator Cloud Portals for Enhanced Efficiency

Although cloud computing applications are only beginning to be adopted by life sciences companies, industry analysts predict that cloud-based, collaborative investigator portals will be used extensively to accelerate clinical trial development, reduce clinical trial study costs, enhance efficiency and improve quality.

In order to facilitate systems integration, enhance investment protection and lower recurring software maintenance costs related to application customization, biopharmaceutical enterprises are increasingly adopting technologies that utilize industry standards. SharePoint lacks native support for RDF/XML, a key standard emerging in the BioPharma data exchange market, and recently adopted by the CDISC/FDA PhUse project and the eTMF Standards initiative for data interoperability.

SureClinical sees collaborative cloud computing for Health Sciences as a means of deploying essential technologies built on open standards and ensuring regulatory compliance. These technologies enable portal-based collaboration across web, mobile and desktop platforms, integrated content management viewing, integrated digital signing technologies, improved business process workflow, centralized reporting, auditing and search, quality management, application customization based on standards, point-and-click integration with existing web-based software applications as well as integrated cloud security that complies with FDA, HIPAA, EU and other regulatory agencies.

The Problems with Microsoft SharePoint

Organizations seeking to use SharePoint 2013 on-premises or cloud version in the highly regulated clinical trial regulatory document area should be concerned with the following issues:

- Cannot complete, E-sign documents entirely in the cloud.
- Not Part 11 compliant, not HIPAA compliant
- Security, governance risk issues [3]
- Requires costly integration to get started and to maintain
- Low user adoption and poor user experience [4],
- Legacy Intranet Portal app designed for use behind firewall [5]

> Of the business managers disappointed with SharePoint, lack of adoption and the user experience were the biggest complaints.
>
> *—Forrester Research Survey, October 2013*

Requirements for Collaborative Cloud Portals for Regulatory Documents

Collaborative cloud portals are used in clinical trial regulatory document management to connect people, organizations, and clinical trial applications and study content in a standards-based, compliant, secure environment. Instead of manually processing clinical trial documents using time-consuming tasks such as printing, signing, and scanning, today's cloud-based applications simplify and automate tasks by enabling users and clinical trial partners to complete documents entirely in the cloud.

Companies are embracing collaborative cloud computing solutions that enable secure information sharing and that eliminate paper through FDA and EU agency compliant digital signatures. Cloud document completion 'beyond the corporate firewall' with business partners fundamentally transforms the way in which clinical trials are conducted. Best-in-class collaborative cloud applications allow users from multiple organizations to

Copyright 2013, 2014 SureClinical Inc.

collaboratively review, digitally sign and complete documents through a centralized, standards-based, collaborative cloud application that is secure, compliant, and automates business processes. The shared cloud portal approach generates economies of scale that reduces costs, accelerates new drug delivery times and facilitates global business organizational agility and flexibility. To realize these benefits, BioPharma organizations are deploying collaborative cloud solutions that connect people, organizations and applications together seamlessly within a compliant, standards-based environment.

Collaborative cloud portals are 'Extraprise' solutions that can do the following:

- Improve collaboration and document exchange capabilities to accelerate the flow of information between sponsors, investigators, research partners and regulators—inside and outside the enterprise firewall.
- Eliminate paper-based processes with cloud-based FDA and EU compliant digital certificate signing to automate workflow and better manage data, providing secure access—anytime, anywhere
- Support standards-based technologies, such as the new eTMF Standard RDF/XML, to facilitate interoperability with existing systems
- Provide a non-modifiable audit trail for digital certificate signing events and user content change actions.
- Comply with Independently audited, certified cloud operations—FDA Part 11, HIPAA, FIPS-140-2-L3, SSAE-16 Type II

The SharePoint Legacy

Today, several eTMF application vendors are promoting SharePoint as the basis for their clinical trial software applications, including Electronic Trial Master File (eTMF) applications and Clinical Trial Management Systems (CTMS). Many vendors are providing solutions built on SharePoint and branding them as cloud eTMFs or cloud CTMSs. The purpose of this white paper is to discuss some of the legitimate considerations that any company should evaluate when contemplating the purchase of a solution that utilizes SharePoint as its underlying architecture. The key factors to assess include ease of use, maintenance and support costs, security, compliance, collaboration and mobile device and cloud support.

In contrast to cloud computing applications which are purpose-built to enable collaboration with multiple organizations, SharePoint was architected in the late 1990s as a general-purpose on-premises workgroup software solution for use behind the enterprise firewall. Owing to its legacy architecture and proprietary schema, SharePoint is unable to adequately address the pivotal factors that are required for success in the cloud. These factors will be addressed further below.

Evaluation of SharePoint as a Collaborative Cloud Platform for Reg Docs

When evaluating SharePoint-based applications for BioPharma, sponsors, CROs or consultants involved in gathering clinical trial source data should closely evaluate the use of SharePoint as the underlying architecture. According to recent surveys, most organizations using or planning to use

SharePoint for enterprise deployments will select the on-premises software deployment option for SharePoint and not the new SharePoint 2013 Online cloud service. For this reason, and because the SharePoint 2013 online version doesn't support digital signing for multiple clinical trial partner organizations outside the firewall, in this evaluation we assume that the SharePoint platform is deployed in a private cloud with secure access by users primarily within the deploying organization.

SharePoint – Difficulties in Sharing Content Outside of the Organization

By default, the on-premises version of SharePoint prohibits users outside your company's network or 'Active Directory' to connect to SharePoint. This makes it difficult for outside stakeholders such as partners, consultants or other clinical trial study team members to securely access documents stored in SharePoint.

There is a workaround to the problem (known as a 'Federated Authentication') but it is rarely used by enterprises because of the complexity, security, administrative and licensing challenges. First, setting up of federated authentication requires that all partner organizations have Microsoft Active Directory installed on their network. Of course, imposing a requirement that a partner own and operate a Microsoft Active Directory computing environment is impractical in most business relationships. Second, in a federated authentication, the primary organization that hosts SharePoint must connect their Microsoft network to the study partners' Microsoft network. To create the connection, a permanent link is established between the firewall of the hosting organization's network and the partner's network. Administration, maintenance of this link and network security policies between the organizations are required in addition to the standard SharePoint user administration and security policies between the organizations.

Evaluation Matrix: Cloud-based Collaborative eTMF Requirements vs. SharePoint

Below are the critical key factors and requirements that need to be considered when deploying collaborative cloud applications for clinical trials.

Standards-based

Requirement	SharePoint 2013 - Enterprise Edition
Web document standards – PDF creation, viewing and archiving	By default, SharePoint supports Microsoft Office formatted documents and OpenXML which are not PDF-compatible and are not web standards. No integrated support for ISO standard PDF document viewing, creation and PDF/A archiving is included with SharePoint. Requires purchase, installation, maintenance of separate PDF components from third-party vendors. No mobile support for PDF viewing through SharePoint.
Digital signing using trusted signing certificates per FDA CFR 21 Part 11 (PKI x.509; FIPS 140-2 L3)	No integrated support for digital signing using certificates. Requires purchase, installation, maintenance of separate certificate signing services, infrastructure and software components from multiple third-party vendors. No mobile support for digital signing.

Standards-based (continued)

Requirement	SharePoint 2013 - Enterprise Edition
eTMF Standard – support for RDF/XML schemas and schema editing	No native support for RDF/XML schemas. SharePoint has a closed schema that does not support direct export/import of RDF/XML schemas. Requires custom development or third-party vendor software.
Export of repository to web standards-based formats for online or offline access	No integrated support for export of content repository metadata, taxonomy and documents. Document download is possible, but options to export a complete eTMF Standard content model with eTMF hierarchy, metadata and documents is not supported.

Compliance

Requirement	SharePoint 2013 - Enterprise Edition
Meeting regulatory compliance requirements; Certifications (FDA, EU)	By default, SharePoint does not offer full FDA compliance and lacks support for key compliance requirements including: -Integrated support for FDA Part 11 electronic signing with x.509 certificates -Integrated audit trail for electronic signature events
Audit trail for electronic signing events	No audit trail for user E-signature events. Third- party solution must be procured, integrated and validated for compliance.
Administrators may not alter electronic signing manifests or audit trails	SharePoint signing events can be tracked in the SharePoint database. However, any user with elevated administrative privileges can modify the SharePoint database and therefore, the audit trail can be modified, compromising security and FDA compliance.
Third-party software components have been independently validated for compliance with FDA Part 11	In order to achieve full functionality as a platform for clinical trial applications, multiple third-party SharePoint components must be added by organizations to their SharePoint implementation. According to FDA Part 11 regulations, all software components must be FDA Part 11 compliant. Because they are added as general aftermarket components, these add-ons like PDF viewers, PDF converters, electronic signature software are not designed for FDA compliance.

Copyright 2013, 2014 SureClinical Inc.

Security

Requirement	SharePoint 2013 - Enterprise Edition
Support for secure access devices using secure networking	SharePoint On-Premises network security is an organization-dependent issue
Non-editable user activity logs and electronic signature event logs	SharePoint users with elevated credentials can modify electronic signature logs or user activity logs that are stored in SharePoint database.

4. Cloud Collaboration

Requirement	SharePoint 2013 - Enterprise Edition
Clinical trial study teams have become global webs of interdependence among sponsors, CROs, investigators, IRB's academic research centers and other study partners. The ability of study team partners to electronically complete, digitally sign and exchange documents in a trusted cloud environment can significantly accelerate clinical trial business results.	Not possible with the default SharePoint platform. True collaboration, document completion and digital signature support between clinical trial team members is impossible using SharePoint unless each business partner owns and operates a Microsoft Active Directory computing environment, as well as their own FDA and EU compliant Esigning infrastructure.
Strong vetting, identification and ongoing authentication of contributors and signing parties to ensure validity of source content, using a third party certificate authority	Third party vetting and identification of clinical trial team members are not included with SharePoint. Without strong identification and external team member authentication, anyone who obtains SharePoint login credentials could fraudulently E-sign documents in the name of another person, invalidating source documents.
Secure collaboration across organization firewalls with support for viewing, completion and E-signatures on mobile, web and desktop devices.	Collaborative cloud document E-signatures using digital certificates are only possible within the SharePoint-sponsoring organization. Collaborative document E-signing across the firewall using FDA compliant x.509 digital certificates with FIPS-140-2 Level 3 compliance is not possible at the time of this publication via SharePoint.
Centralized, auditable service with read-only E-signing and activity audit logs and reporting	SharePoint does not provide any compliance auditing for user activities and for electronic signatures. Auditable compliance transaction logging and monitoring must be provided by the implementing organization.

Ease of use

Requirement	SharePoint 2013 - Enterprise Edition
Easy to use, easy to administer.	Recent independent user surveys by CNET have shown that SharePoint is perceived by the majority of users as difficult to use and to administer. [2]
Ability to work through multiple user interfaces (rich client applications, browser, and mobile devices)while maintaining consistent security and user policies and preferences.	SharePoint supports a web browser interface and has just recently announced support for mobile web browser client support. No rich desktop client.
Easy-to-use organization-specific design of application metadata form fields and eTMF classification taxonomies based on the eTMF Standard RDF/XML format.	No support for the RDF/XML format used by the eTMF Standards initiative and CDISC/FDA is offered by SharePoint.
Easy to manage user access and feature privileges at the feature level	By default, SharePoint lacks the ability to define feature level privileges for users or user groups.

E-sign Clinical Content in Cloud

Requirement	SharePoint 2013 - Enterprise Edition
Integrated support for Electronic signing (Esign) of PDF documents from web, mobile and desktop interfaces with FDA-and-EU compliant digital certificate-based signatures.	No E-signing support
Ability to validate and authenticate electronic signing events	No support
Strong authentication of signers and non-modifiable audit trail of all signing events.	No support
FDA- and EU compliant digital certificate-based signatures.	No

Business Process Management

Requirement	SharePoint 2013 - Enterprise Edition
Support for multiple document business process completion in parallel and serial, with document and medical image capture support from the cloud across multiple organizations, using mobile, web, desktop devices	Not available with SharePoint platform. Must purchase, integrate and maintain third-party solution.
Integrated business process compliance reporting and notifications	Not available with SharePoint platform. Must purchase, integrate and maintain third-party solution.

Copyright 2013, 2014 SureClinical Inc.

Connect to existing clinical trial applications and systems

Requirement	SharePoint 2013 - Enterprise Edition
Provide secure access and visibility into external clinical trial applications and non-Microsoft databases, on-premises or cloud-based through web-standard HTTPS APIs	Not available with the SharePoint platform. Must purchase, integrate and maintain third-party solution. SharePoint schema is closed and only supports reading/writing to Microsoft SQL Server database.

Easy to Extend

Requirement	SharePoint 2013 - Enterprise Edition
Customization of standards-based eTMF classification schema to meet study needs using W3C standards-based tools	Not available with the SharePoint platform. Must purchase, integrate, maintain third-party solution. SharePoint schema is closed and doesn't support modification of the underlying Microsoft SQL Server database schema.

Total Cost of Operation

Requirement	SharePoint 2013 - Enterprise Edition
Cost-effective, easy to finance/purchase	SharePoint total costs are higher than cloud computing costs due to: -Capital costs: Software acquisition requires capitalized purchase and amortization vs. monthly expense of cloud fees.
Low document completion costs	No multi-org collaboration: Inability to provide a centralized collaborative cloud portal that allows multiple organizations to collaboratively review, digitally sign using FDA compliant digital certificates and complete documents for cost savings.
Low maintenance and support costs	Maintenance and support: Higher than cloud solutions due to ongoing maintenance of third-party integrations and third-party plug-in support.
Low ongoing compliance costs	Compliance costs: Continual ongoing compliance costs. As third party vendors modify applications, the entire SharePoint installation must be re-validated.

Why SharePoint Falls Short in BioPharma Extraprise Cloud Collaboration

The SharePoint workgroup collaboration platform was designed and built about 20 years ago in the 1990s. Initially distributed as a software application for use within workgroups behind the firewall on a Windows server, SharePoint has evolved in different directions but has been unable to meet evolving business demands.

While SharePoint's strategy of giving away their licensed software has helped them gain market share in earlier years, the marketplace recently has moved away from the old SharePoint licensed software approach and has turned to cloud computing. The primary reasons that organizations are moving away from SharePoint:

- The move to cloud computing and mobile device growth: SharePoint has only recently announced support for these two critical technologies in SharePoint 2013 and is lagging behind market demand.
- Sub-par user experience: According to recent surveys, a majority of SharePoint users find the user interface lacking and user adoption rates have slowed as a result. [7]
- Administration headaches: A complex user administration interface makes it difficult for administrators to manage SharePoint users. [3]
- High integration maintenance costs: Even basic features like PDF viewing in SharePoint requires integration services, IT support and recurring maintenance of non-Microsoft add-ins. The cost to deploy, validate and maintain an FDA Part 11 compliant SharePoint solution is often higher than purchasing a purpose-built cloud eTMF solution

Given the above factors and considering that the product was designed and still operates like a 1990 intranet file share, it can be concluded that that SharePoint is not ready for the cloud.

SharePoint – Closed, Proprietary, Not Standards-based

From its inception in the 1990s through to the on-premises and cloud versions of SharePoint 2013 today, SharePoint has been and is a Microsoft-proprietary and closed platform. SharePoint has what's known as a 'proprietary database schema,' which means that the way data is stored is not published, making its structure inaccessible by non-Microsoft applications through the ANSI standard SQL database language.

What this means is that integration between SharePoint and external applications and data needs to be done through custom code that is difficult to maintain. In today's collaborative web technology environment, open web standards prevail; not closed, proprietary schemas. The new eTMF standards initiative for clinical trial data exchange uses web standards like RDF/XML and an open database schema to facilitate data sharing. Without support for the creation, editing and sharing of RDF/XML based content with open source tools, SharePoint is not interoperable and will be unable to seamlessly share clinical trial data.

Furthermore, because of its proprietary schema, customers will not be able to easily modify an existing eTMF taxonomy/ontology using SharePoint to meet their organization-specific business needs without heavy custom application custom coding and development, which will require additional maintenance and support costs, and if not done by Microsoft, must be maintained and supported by others.

The limited ability for SharePoint customers to share clinical trial eTMF database and schema using a standard format means that sponsors, CROs and business partners will be unable to conduct automated clinical trial data exchange unless all partners use the SharePoint product.

Central to successful cloud collaboration in clinical trial is the ability to easily find and access the information at any time. This demands a well-defined web-enabled ontology with linkages to internal and external resources, a powerful search experience, flexible navigation, in-line document previewing, clinical trial eTMF Standard metadata – such as classification category annotation and tagging. With a closed schema and a lack of support for emerging clinical standards like RDF/XML, SharePoint's ability to provide extensive and accurate search results for clinical trial data will continue to be hampered. Over time, these key feature failures are likely to lead to further declines in SharePoint user adoption and sustained rates of SharePoint abandonment. [7]

Copyright 2013, 2014 SureClinical Inc.

SharePoint Applications: Patchwork of Custom Point Solutions

Businesses demand robust solutions that provide cloud-based collaboration in clinical trial studies—across mobile, web and desktops channels globally, anytime, anywhere. SharePoint's simple lists and tables are not enough to build complex clinical trial collaborative applications on. As a result, companies are forced to purchase and integrate multiple point solutions and 'SharePoint web parts' to patch together complex custom point solutions in hopes of creating a functional and FDA compliant application.

For instance, to support key features such as PDF document creation and integrated PDF viewing, digital document signing and medical image support, SharePoint customers would need to purchase separate web parts from multiple vendors for each feature. Since very few web part components have completed FDA 21 CFR Part 11 compliance review, it is likely that the entire SharePoint eTMF application is not FDA compliant. While compliance is obviously critical, the ability of an organization to support multiple point solutions for a mission-critical clinical trial eTMF application will be stretched for anything but the smallest clinical trial. In contrast, SureClinical eTMF cloud is a purpose built app that's ready to run from day one and includes the software, services and maintenance needed while ensuring FDA compliance.

The result of this complicated patchwork of SharePoint point solutions is:

- Costly recurring maintenance, management, and administration
- Increased regulatory risk of audit due to FDA 21 CFR Part 11 compliance issues with SharePoint
- Every new feature request turns into a major system integration project
- Excessive consulting and services costs, as much as 10x the original license costs.
- No support for new RDF/XML standards being adopted by CDISC/FDA, eTMF Standard
- A high risk technology path that doesn't protect IT investments in the clinical trials area, and leaves customers to shoulder the compliance and technology risks

Microsoft claims its new SharePoint 2013 online service will enable collaboration. However, its not FDA or HIPAA compliant, and lacks integrated digital signing and business process management services that are requisite features in the elimination of paper-based manual processes in extraprise collaborative computing.

Sureclinical's Approach – What Sets Us Apart

The collaborative cloud computing approach is gaining significant traction in enterprises looking to accelerate business results, and reduce operating costs. BioPharma organizations of all sizes are finding that portal-based collaborative cloud computing applications that are FDA and EU compliant deliver more value at a lower cost and with faster time-to-value than premises-based or single-enterprise alternatives such as SharePoint. Companies are selecting SureClinical today for the following reasons:

- The ability of study team partners to electronically complete, digitally sign and exchange documents in a trusted FDA and HIPAA compliant cloud environment
- Faster clinical study startup with easy-to-use and easy-to-configure solution
- Lower total application operation, maintenance and support costs
- Reduced overall study costs through the elimination of paper-based processes
- Compliance, security and cloud certifications: FDA Part 11, HIPAA, EU, SSAE-16 Type II, FIPS-140-2-L3
- Standards-based

Copyright 2013, 2014 SureClinical Inc.

An end-to-end cloud application solution

SureClinical eTMF cloud is a purpose-built cloud application designed to support collaboration among a trusted network of clinical study partners. SureClinical eTMF is an end-to-end cloud application solution with integrated capabilities that span clinical trial electronic trial master file content management, high-trust digital signature support, document completion workflow automation, source document quality control and contributor identity validation tools, and integration with existing on-premises and cloud-based clinical trial applications.

Operational efficiency

With SharePoint-based eTMF solutions, enterprises need to purchase, license and maintain SharePoint as well as an additional eTMF application. Then, if additional services are needed for digital signatures, these would need to be procured, managed and maintained separately. With SureClinical eTMF, study managers can purchase a single cloud application solution, eliminating the need to purchase and maintain multiple applications or expensive add-on point solution licenses or signing infrastructure from third parties. This also means that you avoid the resulting excessive software installation, system integration, management, administration, maintenance costs and compliance issues.

Simple, responsive interface

With SureClinical eTMF, clinical trial users can use a web, mobile or desktop interface to view and manage clinical trial documents, using an intuitive and easy-to-learn user interface.

SureClinical eTMF cloud has a broad set of capabilities that help you accelerate document workflows with minimal support. In most cases, new clinical trials can get started with SureClinical in under a week with our cloud-based system. Study teams administer groups of clinical study users and partners through a simple-to-use administrative interface.

Highly scalable

SureClinical eTMF cloud can scale to the largest enterprises and extraprise networks across multiple clinical trials: With SureClinical eTMF cloud, your deployment can support millions of documents and content items across hundreds of investigators, partners, and CROs, as well as thousands of users distributed worldwide. SureClinical provides a clear path for growth away from costly on-premises software installation, management and maintenance to cloud collaboration environment to meet enterprise-wide needs.

Interoperability

By using an open architecture which supports multiple database sources and standards, there are optional interfaces available to enhance and extend the SureClinical eTMF platform. Your team can use open source or SureClinical's SureDesigner tool to do so. Since SureClinical eTMF is based on the Oracle® Java® platform and includes an optional web API, a wide variety of on-premises and cloud-based applications can be integrated.

Configurable Using Web Standards

Using the SureDesigner content model, power-users can easily customize clinical trial study portal content using standards-based formats such as the new eTMF Standard for interoperability, RDF/XML, enabling content to be shared and re-used with other clinical studies and even with other applications. The use of drag-and-drop editing tools lets users streamline business operations with an intuitive user interface.

Copyright 2013, 2014 SureClinical Inc.

Easy user adoption

Your organization can benefit by easily finding trained consultants and quickly getting them up-to-speed and productive with a solution that can be activated in minutes, not months..

SureClinical eTMF cloud is easy for end-users to learn and includes quick-start wizards and user training that helps end-users to be self-sufficient. Simple to use end-user settings enable self-provisioning capabilities, personal privacy settings, and the ability to modify profiles.

SureClinical eTMF cloud is designed to facilitate user adoption and usage by providing a familiar and contextual user interface across web, mobile and desktop devices. Additionally, SureClinical eTMF is the only eTMF application that delivers integrated document viewing and navigation with its SureClinical Triple-View™, which allows inline PDF viewing along with the eTMF taxonomy and tags, providing a superior experience that enhances productivity for users anytime, anywhere.

SureClinical eTMF cloud delivers a great mobile and offline experience. Using patent-pending mobile digital signature and document viewing technologies, the SureClinical eTMF mobile application provides secure access of clinical trial documents for any trusted and pre-authorized study member. Clinical research associates and other study team members can view, search and share documents, medical images and reports from within their mobile device browser.

SureClinical eTMF balances this end-user empowerment with the necessary business control by providing granular security and privacy policies, centralized management, and reporting and analysis capabilities.

Conclusion

The benefits of cloud collaboration for clinical trials over SharePoint are clear and are rapidly being adopted by multiple organizations today.

SureClinical and the SureClinical eTMF cloud application support your organization’s move to the cloud with the industry’s first 100% paperless collaborative cloud solution for clinical trial study sponsors, CROs, investigators and partners.

SureClinical eTMF is the first comprehensive single-vendor solution that enables study participants to digitally sign documents anytime, anywhere, reducing the need for paper, minimizing manual paper handling costs and accelerating the delivery of new drugs to the market. It changes the way clinical trials will be conducted in the cloud by helping organizations to quickly build networks of trusted clinical trial partners.

Using a standards-based, purpose-built application with FDA and EU compliance, SureClinical eTMF includes optional integrated digital signature support as well as workflow, document quality and medical image management. Support for integration with on-premises and cloud-based external applications through web services is also available.

It delivers an optimized user experience across desktop, browser, and mobile devices, increasing clinical trial user productivity and reducing study time and time to market. Users benefit from the best-in-class collaboration experience provided in simple-to-use mobile, web and desktop interfaces.

Copyright 2013, 2014 SureClinical Inc.

Get Started Today With SureClinical

SureClinical eTMF delivers a standards-based architecture with application customization, that is easy to extend and requires zero maintenance. The integrated offering of key features—multi-organization cloud collaboration, digital certificate signature support, automated content workflows, quality management, medical image management, advanced search and reporting and compliance—enables study team collaboration scenarios using clinical trial best practices. SureClinical eTMF, as the premium collaborative cloud application vendor in the health science content management area, has a demonstrated track record of industry leadership, innovation and continued investment that you can rely on.

References and Resources

This paper refers to the following sources:

[1] InfoTrends, “SharePoint 2010 Cost of Ownership: Expect the Unexpected,” December 2011

https://www.oracle.com/us/products/middleware/webcenter/sharepoint-cost-of-ownership-1508591.pdf

[2] ZDNet, SharePoint: ‘Can it navigate the cloud, mobile curves ahead?,’ October 2013

http://www.zdnet.com/sharepoint-can-it-navigate-the-cloud-mobile-curves-ahead-7000022162/

[3] CIO Insight, ‘The Problems With Microsoft SharePoint,’ March 2013

http://www.cioinsight.com/it-strategy/messaging-collaboration/slideshows/the-problems-with-microsoft-sharepoint-14/

[4] PCWorld, ‘Microsoft SharePoint faces a challenging future: Forrester,’ Feb 2013

http://www.pcworld.com/article/2027391/microsoft-sharepoint-faces-a-challenging-future-forrester.html

[5] CiteWorld, ‘How did SharePoint become the poster child for bad enterprise software?,’ Jan 2013

http://www.citeworld.com/social/21352/how-did-sharepoint-become-product-everybody-loves-bash?page=0

[6] Stephen E. Arnold: ‘Business Process Management missing in SharePoint,’ October 2013

http://www.phibetaiota.net/2013/10/stephen-e-arnold-open-source-at-oracle-and-with-sharepoint/

[7] AIIM Survey, ‘SharePoint 2013, Clouding the Issues, October 2013

http://goo.gl/VNCBry

SureClinical Inc.

2865 Sunrise Blvd Suite 220
Rancho Cordova, CA 95742
Tel: 916-265-2000
info@SureClinical.com
www.SureClinical.com

Dec 2014